

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460674.961 +/- 0.029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.029
Transit depth (Rp/Rs)^2	2.27 +/- 0.87 [%]
Orbital Inclination (inc)	83.82 +/- 2.25
Airmass coefficient 1 (a1)	1.2531 +/- 0.006
Airmass coefficient 2 (a2)	-0.178 +/- 0.043
Scatter in the residuals of the lightcurve fit is	0.48 %
Best Comparison Star	None
Optimal Aperture	3.83
Optimal Annulus	11.91
Transit Duration (day)	0.045 +/- 0.04

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.029 vs 0.1276 (deviation 0.81σ)
Duration fit vs published	0.045 d vs 0.1179 d (deviation 1.82σ)
Mid-time O–C	-41.7 min
Depth SNR	5.2
Residual scatter	0.48 %

Light curve

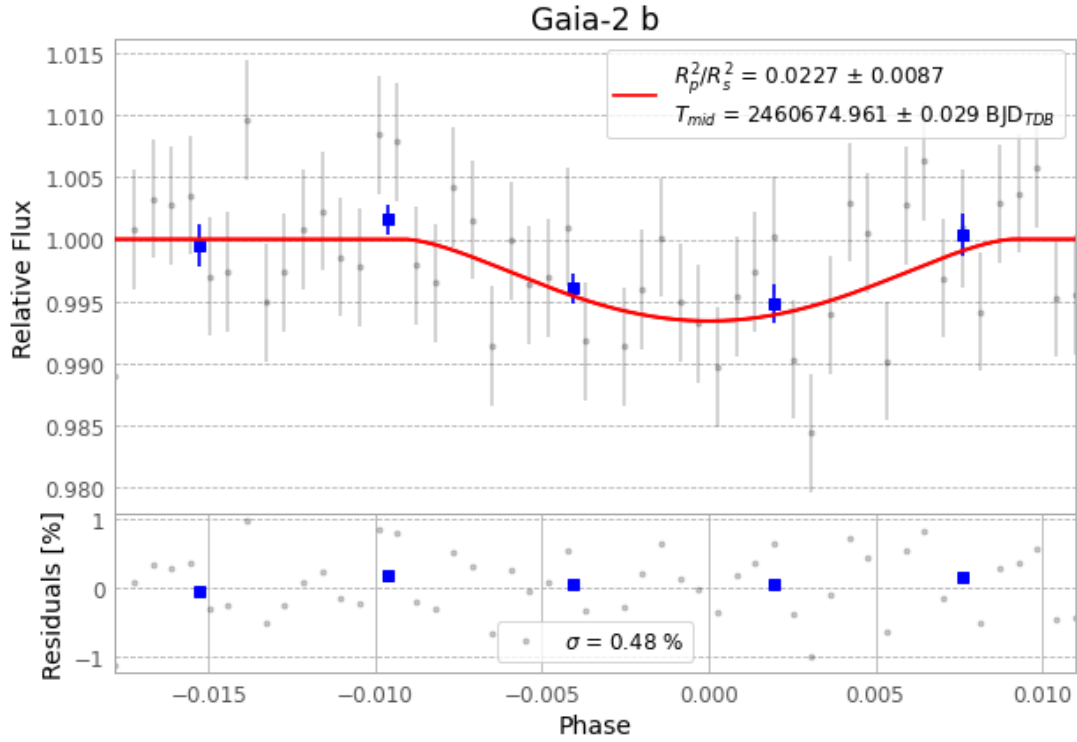


Figure 1 — FinalLightCurve_Gaia-2 b_2024-12-30.png (SHA-256 672cb5c62973985a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 286], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 484de5e9200c5e10...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

52 FITS frames (34 MB), Gaia-2241230092115.FITS → Gaia-2241230115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Gaia-2-241230.log (0f4288c6a943...), FinalLightCurve_Gaia-2 b_2024-12-30.png (672cb5c62973...), FinalLightCurve_Gaia-2 b_2024-12-30.csv (32e77eaba0ac...)

Compute resources

Wall time 128.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 22:48Z.